Length of an arc on the surface

Def. Let X be a patch and

$$\alpha: [a,b] \to \mathbb{R}; t \mapsto X(u(t), v(t)),$$

then the arc length is

$$L(\alpha) = \int_a^b |\alpha'(t)|\, dt = \int_a^b \sqrt{I_{\alpha(t)}(\alpha'(t))}\, dt$$

Area of Torus

$$\text{Area}(X(B)) = \iint_B \sqrt{EG - F^2}\, du\, dv.$$

Consider the torus : for $0 < \phi, \theta < 2\pi$,

$$X(\phi, \theta) = \big((a + r\cdot\cos\theta)\cdot\cos\phi,\ (a + r\cdot\cos\theta)\cdot\sin\phi,\ r\cdot\sin\theta\big)$$

So that

$$X_\phi = \big(-(a + r\cos\theta)\sin\phi,\ (a + r\cdot\cos\theta)\cdot\cos\phi,\ 0\big)$$

$$X_\theta = \big(-r\sin\theta\cos\phi,\ -r\sin\theta\sin\phi,\ r\cos\theta\big)$$

Now, $E = (a + r\cdot\cos\theta)^2$,

$F = 0$

$G = r^2$.

Now,

$$\text{Area} = \lim_{\varepsilon \to 0} \int_{\varepsilon}^{2\pi - \varepsilon} \int_{\varepsilon}^{2\pi - \varepsilon} ra + r^2\cos\theta \; d\phi\, d\theta.$$